A SPACE IS HAUSDORFF EXACTLY WHEN ITS DIAGONAL IS CLOSED IN THE PRODUCT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The Hausdorff separation axiom and the product topology on the square, with the diagonal as the set of pairs with equal entries. Set up in the author's section on Hausdorff spaces, immediately before the counted statement. Fixes the two conditions whose equivalence the statement asserts.

Lemma 0.1. *Let X be a topological space. The following are equivalent:*

(1) *X is Hausdorff,*
(2) *the diagonal $\Delta(X) \subset X \times X$ is closed.*

Proof. We suppose that X is Hausdorff. Let $(x, y) \notin \Delta(X)$, i.e., $x \neq y$. There are U and V disjoint open sets of X such that $x \in U$ and $y \in V$. This implies that $U \times V \subset (X \times X) \setminus \Delta(X)$. This shows that $(X \times X) \setminus \Delta(X)$ is an open set of $X \times X$ which is equivalent to saying that the diagonal $\Delta(X) \subset X \times X$ is closed in $X \times X$. The converse is similar: The complement $(X \times X) \setminus \Delta(X)$ consists precisely of $(x, y) \in X \times X$ with $x \neq y$. Thus, if $\Delta(X) \subset X \times X$ is closed, then, by the definition of the product topology, for every such (x, y), there are opens $U, V \subset X$ with $(x, y) \in U \times V$ and $(U \times V) \cap \Delta(X) = \emptyset$. In other words, with $x \in U$ and $y \in V$ such that $U \cap V = \emptyset$. □

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